



EXECUTIVE SUMMARY

Technology and Business Leader with 24+ years of experience building intelligent multi-disciplinary systems combining electronics, embedded software, sensing technologies, and data platforms across automotive, industrial, semiconductor, consumer, defence, and healthcare domains.

Pioneering the use of **AI-augmented engineering** to accelerate the design of hardware, software, and analytical systems. Brings a rare blend of deep technical architecture, product development, and business leadership spanning product strategy, engineering execution, and market development.

Known for architecting complex, scalable solutions and building high-performance deep-tech teams, with hands-on expertise in system architecture, product management, cross-functional engineering execution, and end-to-end business leadership across marketing, sales, and P&L management.

AI-AUGMENTED SYSTEM ARCHITECTURE & ENGINEERING

AI-Augmented Engineering	Leverage modern AI tools to accelerate system modelling, multi-physics simulation, reliability analysis, software development, statistical and financial analysis.
Systems Architecture Vision	Define scalable architectures for complex multi-disciplinary products combining electronics, embedded systems, software platforms, and intelligent algorithms.
Rapid Architecture Prototyping	Use AI-assisted workflows to quickly explore design alternatives, validate concepts, and accelerate early system prototyping.
Intelligent IoT System Design	Design connected systems integrating sensors, control systems, edge computing, and cloud intelligence for real-world environments.

DEEP-TECH SYSTEMS & TECHNICAL EXPERTISE

Multi-Disciplinary Product Design	Develop integrated engineering solutions spanning sensors, instrumentation, control systems, embedded firmware, application software, and cloud platforms.
Sensor, Control & Edge Systems	Architect real-time embedded systems for sensing, control, data acquisition, signal processing and edge intelligence.
Hardware Engineering Depth	Hands-on expertise across mixed signal design, power electronics, FPGA platforms, signal integrity, and EMI/EMC considerations across domains.

PRODUCT, LEADERSHIP & BUSINESS EXECUTION

End-to-End Product Leadership	Drive the full lifecycle from concept and architecture through development, launch, and continuous product evolution.
Cross-Functional Engineering Leadership	Lead multidisciplinary teams spanning hardware, firmware, application software, cloud platforms, manufacturing, and QA.
Deep-Tech Team Building	Recruit, mentor, and scale high-performance engineering teams focused on complex product development.
Design Governance & Quality	Establish architecture standards, design reviews, and reusable engineering patterns for reliable delivery.
Technology-Driven Business Strategy	Align system architecture and product decisions with cost structures, pricing models, and long-term business strategy.
Customer Insight Translation	Convert complex domain needs into clear product roadmaps, engineering priorities, and market-aligned solutions.

ADVISORY & CONSULTANCY INTERESTS

AI-augmented engineering, Edge AI architecture, IoT platform scalability, Intelligent connected products, multi-disciplinary hardware-software platforms, techno-business strategy, product strategy.

EXPERIENCE SUMMARY & SIGNATURE PROJECTS

Role	Company	Projects	Responsibilities
Founder & Architect: AI-Augmented Systems	Anvation Labs 2025 – Present (1+ year)	- Defence: Drone swarm AI PoC - Industrial: Robotic Dish Washer MVP - Industrial: Electromagnetic Flow Transmitter	AI-augmented system architecture, product strategy, advanced prototyping
CEO & Chief of Product Strategy	Anvation Labs 2020 to 2025 (5 yrs)	- Automotive: 2W / 3W Smart Digital Dashboard - Automotive: Self-balancing 2W Vehicle Control Unit - Defence: Night Vision Surveillance Drone - Defence: Laser Directed-Energy Weapon Controller - Defence: HW Secure Crypto Device (IDEX)	Product strategy, architecture definition, engineering leadership, and business growth
Founder & Chief System Architect	Anvation Labs 2015 to 2020 (5 yrs)	- Healthcare: Portable ICU Health Monitoring System - Healthcare: Smart Health Monitoring Wrist Band - Consumer: Bluetooth Enabled Smart Helmet - Consumer: Gesture based Game Controller - Consumer: CV & XR based Smart Fashion Mirror	Product management, architecture definition, engineering execution & delivery
Product Architect	Honeywell 2013 to 2015 (2.3 yrs)	- Industrial: UL & CSA grade Slim Industrial Data Recorder - Industrial: SIL grade PLC Power Supply Platform	Product architecture, system design, and cross-discipline engineering execution.
Director, BD & Engineering	Udupi Tech 2012 to 2013 (~1 yr)	Healthcare: Automatic Hospital Bed Controller System and associated solution	Engineering delivery, customer engagement, and technical business development.
Co-Founder & BU Head	Sankalp Semi 2006 to 2012 (6 yrs)	- Semicon: Full-Chip Analog IC Modelling Framework - Semicon: Post-Silicon Characterization of Mixed-Signal ICs - Semicon: Digital IO & Standard Cell Libraries (32nm) - Semicon: VLSI Design, Layout & Verification Services	Semiconductor engineering leadership, project delivery, and team development.
Co-Founder & CTO	Eklakshya 2006 to 2008 (2 yrs)	Semicon: Industry-aligned VLSI design training and fellowship programs for engineering graduates.	Program design, training delivery, and operations management.
Senior Research Scientist	Sci MEMS Lab 2004 to 2006 (2.5 yrs)	- Defence: Non-Linear Controller for Missile Actuator - Defence: MEMS Chip for Capacitive Sensing - Defence: Piezo Chip for Load Sensing	Advanced research, system modelling, architecture design, and experimental validation.
Scientist B	RCI, DRDO 2001 to 2002 (1 yr)	Defence: S2A / W2A Missile Propulsion & Guidance Control Systems. Test infrastructure and automation for propulsion system validation.	Research engineering, experimental systems development, and test automation.

ACHIEVEMENTS & EDUCATION

- (Co)Founder** of [Anvation Labs](#) (Embedded Products), [Sankalp Semi](#) (VLSI Services), [EkLakshya](#) (VLSI Training)
- 2 Patents** ([EMI/EMC](#), [UX](#)) & **3 Trade Secrets** within 2yrs during Honeywell tenure
- ME in Microelectronics & Product Design from CEDT, IISc, Bangalore (CGPA 7.6/8) – **Gold Medallist** in 2004
- AIR (All India Rank) 5** with 99.94 percentile in GATE 2001 & BE in ECE from NITK in 2001 (77%)
- Trainer** for Program Management, Product Design, Analog Design and Costing Design
- Multiple major awards throughout the career for Innovation, Leadership & Outstanding Achievements